

AI/ML intern DAY 19

AI/ML Internship Final Project Reportp

Research and Analysis of AI-Based Virtual Try-On and Fashion Recommendation Systems

LOYA PRANAY (pranay8679@gmail.com)

AI/ML Intern

Date: 05-06-2026

Certificate / Declaration

Declaration

I hereby declare that this report entitled "**Research and Analysis of AI-Based Virtual Try-On and Fashion Recommendation Systems**" is the result of the work carried out during my AI/ML Internship at Raritone.

The report represents my understanding, research, and implementation activities performed throughout the internship period.

Acknowledgement

I sincerely thank the Raritone team for providing me with the opportunity to work as an AI/ML Intern.

This internship helped me understand the practical applications of Artificial Intelligence, Computer Vision, Machine Learning, and Virtual Try-On technologies used in modern fashion platforms.

I would also like to thank my mentors and teammates for their guidance and support throughout the internship.

Contents

1.Introduction3

2. Project Objectives3

3. AI Development Environment.....3

4. Computer Vision Fundamentals4

5. Body Detection and Pose Estimation.....5

6. Body Measurement Extraction6

7. Fashion Datasets7

8. AI-Based Virtual Try-On Systems7

9. Garment Warping and Fitting8

10. 3D Avatar Generation9

11. AI Recommendation Systems10

12. Complete AI Workflow10

13. Challenges and Solutions12

14. Future Improvements12

15. Conclusion12

1. Introduction

The rapid growth of Artificial Intelligence has transformed the fashion industry by introducing intelligent Virtual Try-On systems and personalized recommendation engines.

Modern online shopping platforms aim to reduce product returns and improve customer satisfaction by allowing users to digitally visualize garments before purchase.

The primary objective of this internship was to research the technologies required for building a complete AI-powered Virtual Try-On platform.

The work involved studying computer vision models, body segmentation techniques, pose estimation systems, garment warping methods, 3D avatar generation, recommendation engines, and fashion datasets.

2. Project Objectives

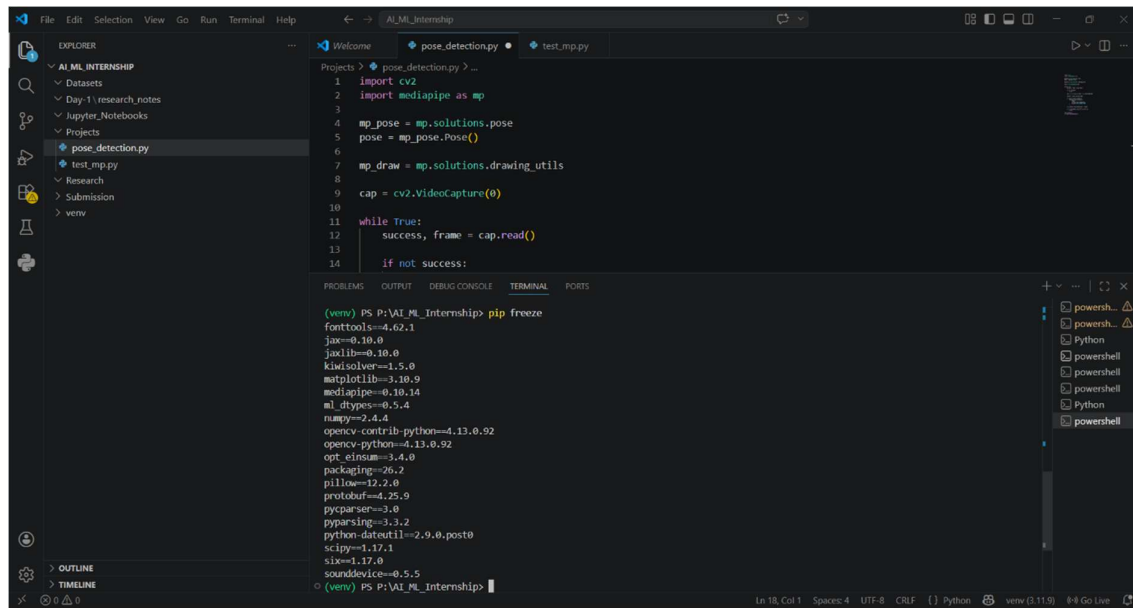
The major objectives were:

- Study Computer Vision fundamentals.
 - Understand human pose estimation.
 - Research body measurement extraction.
 - Analyse AI fashion datasets.
 - Explore 2D and 3D Virtual Try-On systems.
 - Study garment warping techniques.
 - Research AI recommendation systems.
 - Understand complete AI workflows.
-

3. AI Development Environment

Tools Installed

Tool	Purpose
Python	Programming
OpenCV	Image Processing
MediaPipe	Pose Estimation
TensorFlow	Deep Learning
VS Code	Development
GitHub	Version Control



4. Computer Vision Fundamentals

Topics studied:

- Image Processing
- Human Detection
- Landmark Detection
- Pose Estimation
- Skeletal Tracking

Applications:

- Fitness Tracking
- Healthcare
- Motion Capture
- Virtual Try-On

5. Body Detection and Pose Estimation

MediaPipe Pose

Advantages:

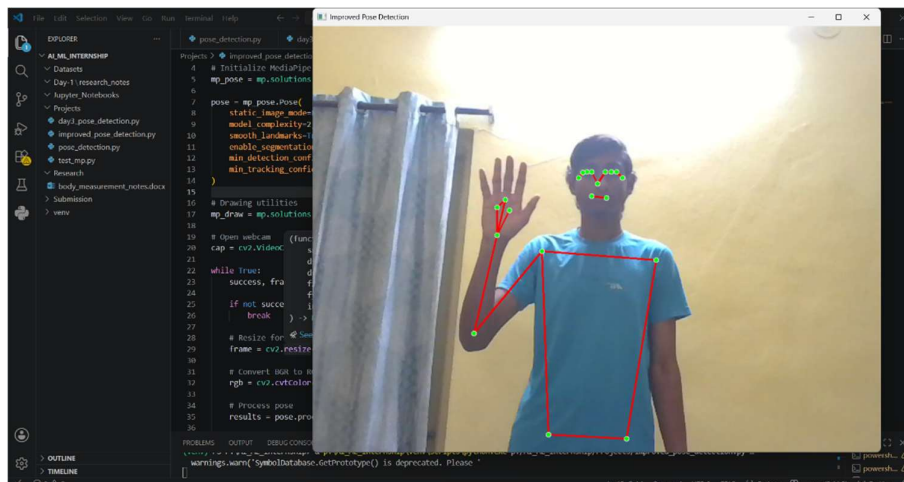
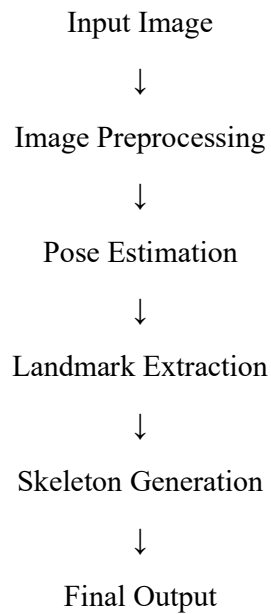
- Real-time processing
- Lightweight
- 33 landmarks

OpenPose

Advantages:

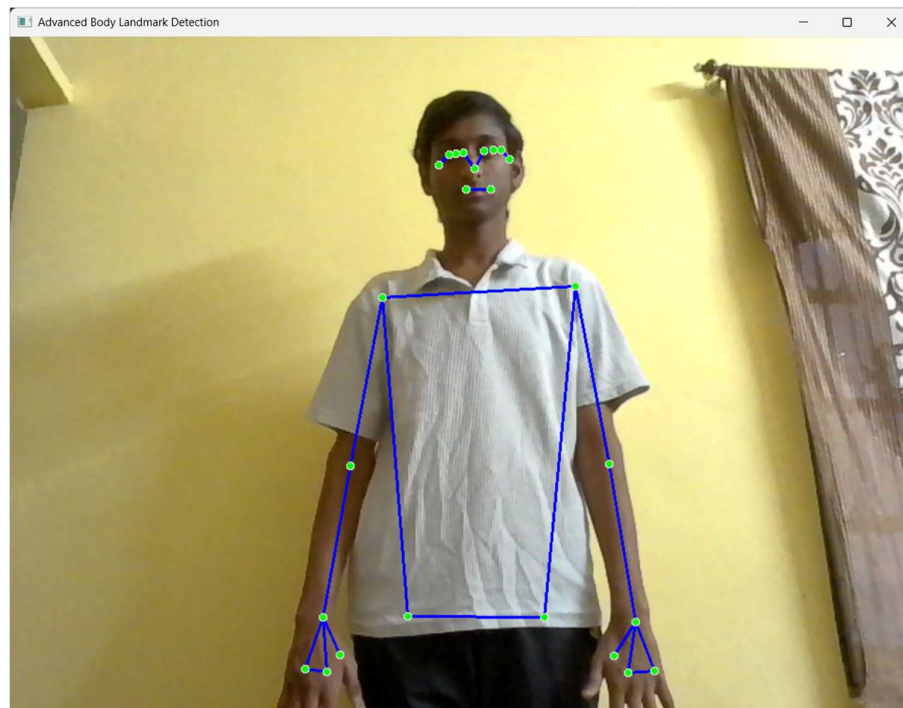
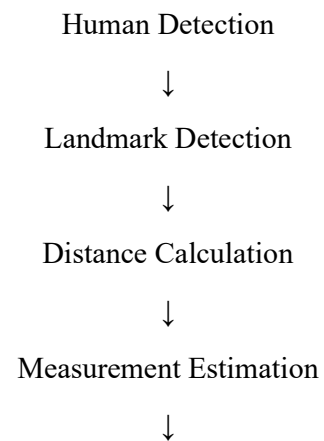
- Multi-person support
- Higher accuracy

Include your Body Detection Workflow:



6. Body Measurement Extraction

Workflow:



Output



7. Fashion Datasets

Datasets Studied

Dataset	Primary Purpose	Key Features
DeepFashion	Fashion Recognition	Clothing categories, attributes, landmarks
DeepFashion2	Clothing Detection & Segmentation	Bounding boxes, pose, segmentation masks
VITON	2D Virtual Try-On	Person-cloth image pairs
VITON-HD	High-Resolution Virtual Try-On	Improved image quality and texture preservation
Dress Code	Multi-Category Try-On	Multiple clothing categories with diverse poses
MPV Dataset	Multi-Pose Virtual Try-On	Various body orientations and garment combinations
Fashionpedia	Fashion Attribute Recognition	Rich semantic annotations and garment attributes

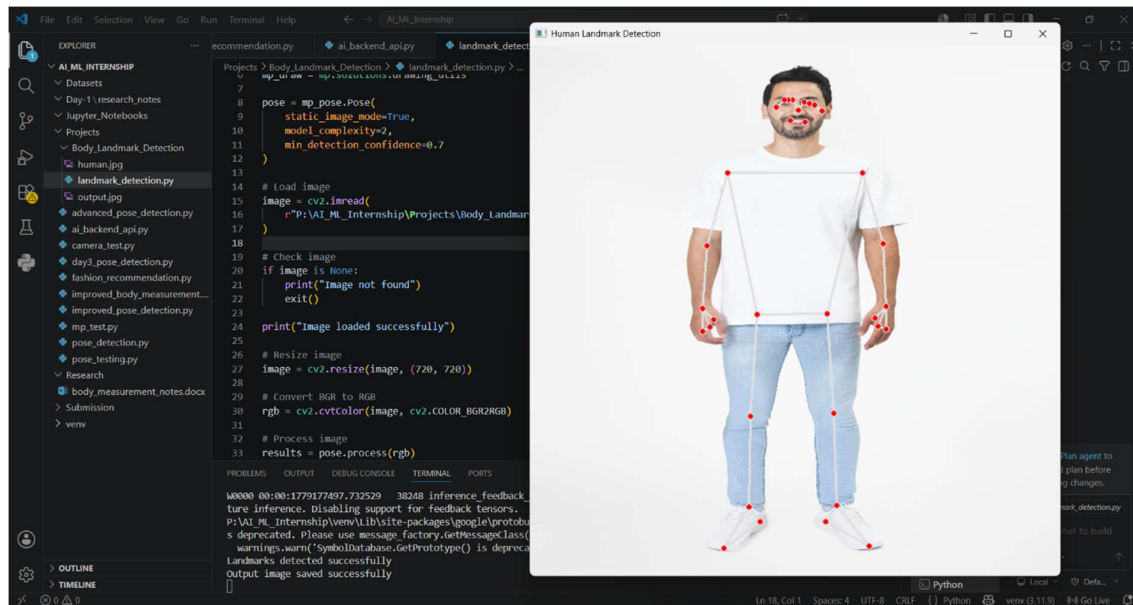
8. AI-Based Virtual Try-On Systems

Models Explored:

- VITON
- CP-VTON
- HR-VITON
- StableVITON
- IDM-VTON

Comparison Table:

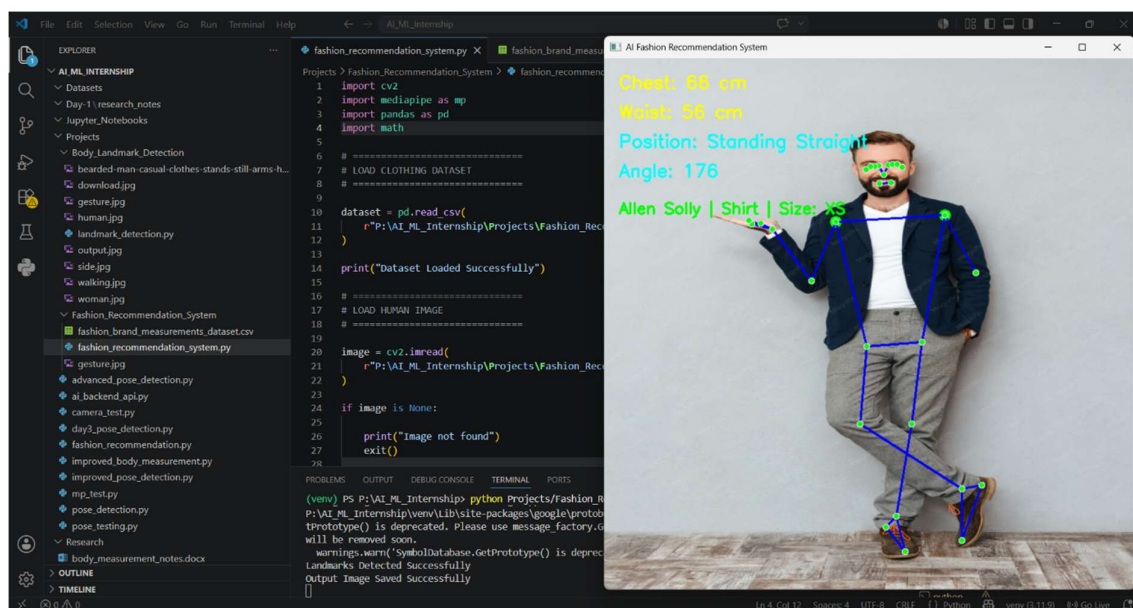
Model	Strength
VITON	Simple
CP-VTON	Better fitting
HR-VITON	High Resolution
StableVITON	Diffusion Based
IDM-VTON	Identity Preservation



9. Garment Warping and Fitting

Topics:

- TPS
- GMM
- Flow-Based Warping
- Affine Transform



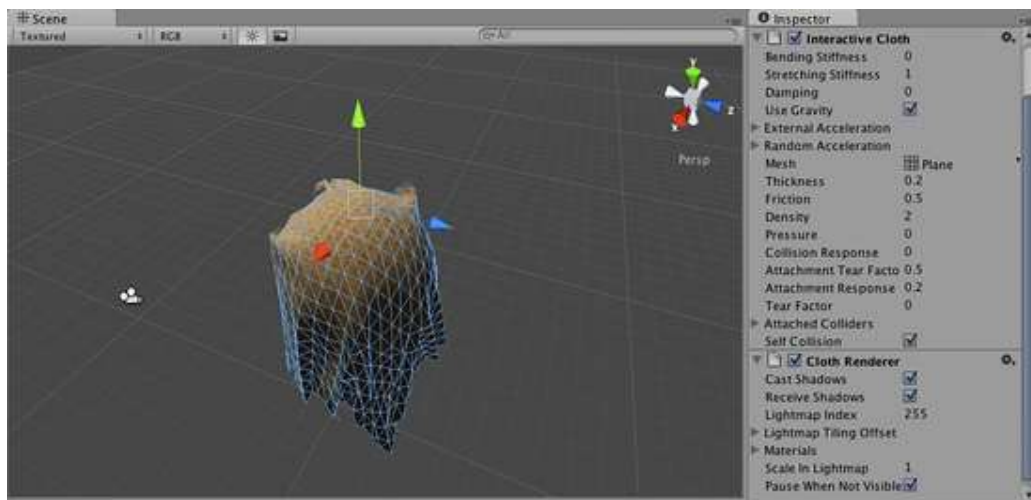
10. 3D Avatar Generation

Models:

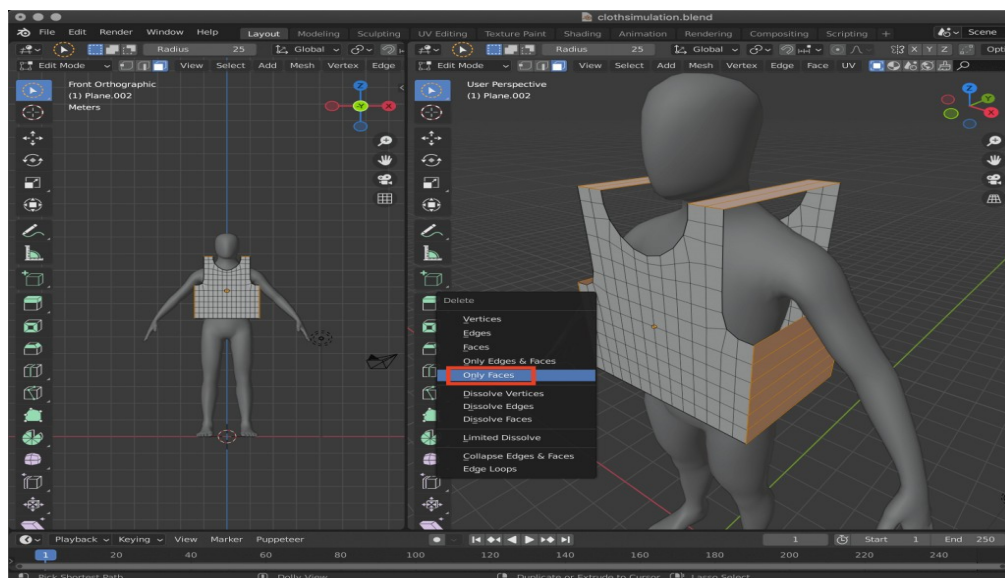
- SMPL
- PIFuHD
- ICON
- HumanNeRF

Cloth Simulation:

- Unity Cloth
- NVIDIA PhysX
- Marvelous Designer
- Blender Cloth



Unity Cloth



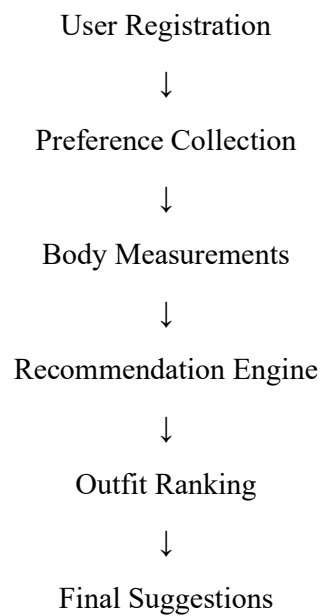
Blender Cloth

11. AI Recommendation Systems

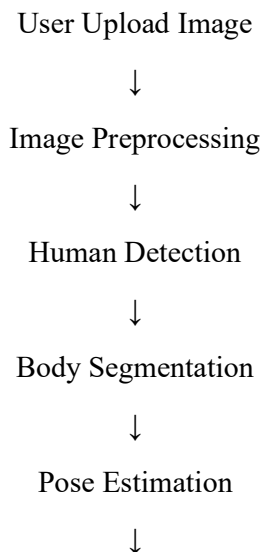
Research:

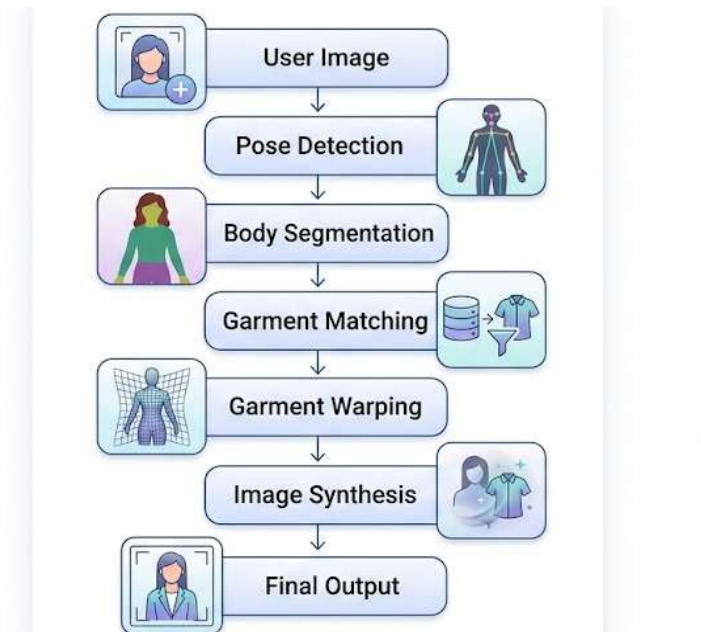
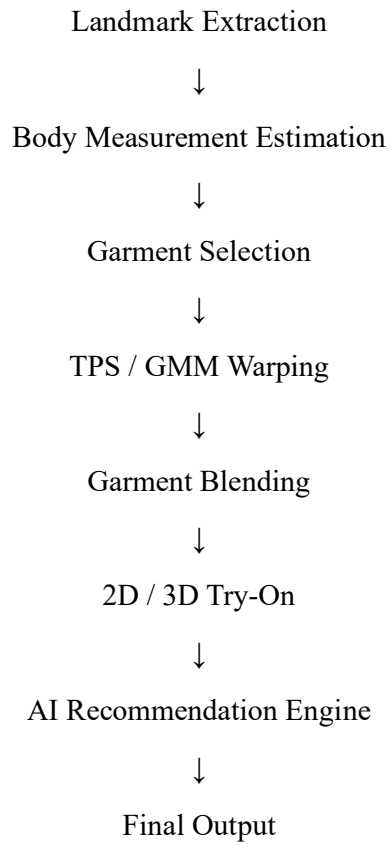
- Content-Based Filtering
- Collaborative Filtering
- Hybrid Systems
- Deep Learning Recommendations

Proposed Workflow:



12. Complete AI Workflow





13. Challenges and Solutions

Challenge	Solution
Landmark flickering	Smoothing
Garment misalignment	TPS
Background leakage	Segmentation
Floating garments	Alpha blending
Complex poses	OpenPose
Cloth deformation	GMM
Dataset inconsistency	Data cleaning

14. Future Improvements

- HR-VITON implementation
 - Stable Diffusion integration
 - Real-time mobile application
 - Full 3D avatar support
 - Multi-angle try-on
 - Personalized recommendations
 - Large-scale dataset expansion
-

15. Conclusion

This internship provided a comprehensive understanding of Artificial Intelligence applications in the fashion industry, particularly in Virtual Try-On and recommendation systems.

The research explored body detection, pose estimation, body measurement extraction, fashion datasets, garment warping, cloth simulation, and both 2D and 3D Virtual Try-On technologies. Various state-of-the-art models including MediaPipe, TPS, GMM, HR-VITON, SMPL, and PIFuHD were studied to understand their role in creating realistic digital fashion experiences.

The internship also emphasized the importance of computer vision, deep learning, and high-quality datasets in building scalable AI-powered fashion platforms. The knowledge gained through this work provides a strong foundation for future research and development in AI-driven e-commerce and intelligent virtual fitting systems.
